Leo Bogaert

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EDUCATION

Western University **Apr. 2029**
Bachelor of Engineering Science, Electrical Engineering (B.E.Sc.) *London, ON*
- GPA: **3.9/4.0**; Dean's Honour List

HARDWARE PROJECTS

Custom Class D Amplifier
- Designed a Class D audio amplifier PCB to drive a 4 Ω speaker using a MOSFET switching stage achieving ~80% efficiency
- Utilized a 555 timer and comparator converting analog audio into a digital PWM signal for the amplifier's MOSFET stage
- Designed the audio signal path, output filtering, and power-supply decoupling for clear and reliable audio reproduction while selecting mass-produced components based on voltage, current, and switching requirements
- Developed and prepared a 2-layer PCB in Altium Designer for fabrication. The design followed all JLCPCB manufacturing rules used DRC with custom created rules for verification

STM32-Based Flight Controller Board
- Designed a flight controller PCB based around an STM32F446, with IMU, barometer, USB, SWD programming, and nine PWM output modules. Sensors and outputs were customized to our needs in an RC fixed-wing aircraft
- Configured a 4-layer stack-up and routed USB differential pairs to a controlled impedance of approximately 90 Ω while applying grounding, return-path, and power-distribution considerations
- Utilized SPI/I2C, 8 MHz crystal oscillator, and integrated 3.3 V/5 V regulation to power and communicate on the board

3D Printed RC Drone
- Developed a high-strength, rigid drone frame using Fusion 360, printed out of carbon-filled PETG to ensure minimal vibrations
- Performed iterative flight testing to validate structural rigidity and tune control responsiveness using Betaflight to achieve stable, responsive flight performance
- Interfaced an RC receiver with the flight controller over UART, enabling wireless command and control

EXPERIENCE

Western WEBOTS **Sep. 2025 – Apr. 2026**
Low Voltage Electrical & Mechanical Member *London, ON*
- Designed a custom harmonic drive with a 20:1 reduction ratio, implemented as a hip joint for humanoid robots
- Designed a central controller utilizing a NVIDIA Jetson Orin Nano to run the autonomous navigation systems
- Configured a high-speed brushless motor driver to enable movement of robot joints

Mandarin **Apr. 2023 – Sep 2025**
Host *London, ON*
- Built advanced communication skills, consistently conveying information clearly between customers and management
- Assisted management by training new employees in phone operations, dishes and hosting procedures

Enders Direct **Sep. 2023 - Feb 2024**
Computer Hardware Technician *London, ON*
- Studied under a senior technician, developing practical skills in troubleshooting and repair methodology
- Built a strong competency in identifying and resolving computer hardware faults and software-related issues

Technical Skills

- **Electronics & PCB:** Altium Designer, KiCad, Analog/Digital/Power Circuits
- **Embedded & Communication:** ESP32, STM32, UART, SPI, I2C, PWM
- **Programming:** C++, C#, Python, Java
- **CAD & Development:** Fusion 360, OnShape, Git, GitHub, Bambu/Prusa Slicer
- **Lab & Debugging:** Oscilloscope, Digital Multimeter, Function Generator, Soldering, Serial Debugging